

Dc-dc converters are power electronic circuits that convert a dc voltage to a different dc voltage level, often providing a regulated output.

LINEAR VOLTAGE REGULATORS

This type of circuit is called a **linear dc-dc converter** or a **linear regulator** because the transistor operates in the **linear region**, rather than in the saturation or cutoff regions. One method of converting a dc voltage to a lower dc voltage is a simple circuit as shown in Fig. 6-1. The output voltage is

$$V_O = I_L R_L$$

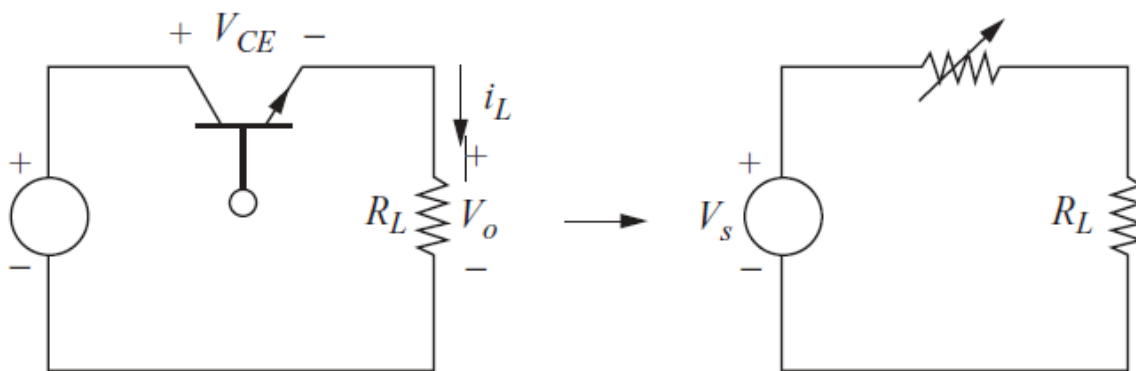


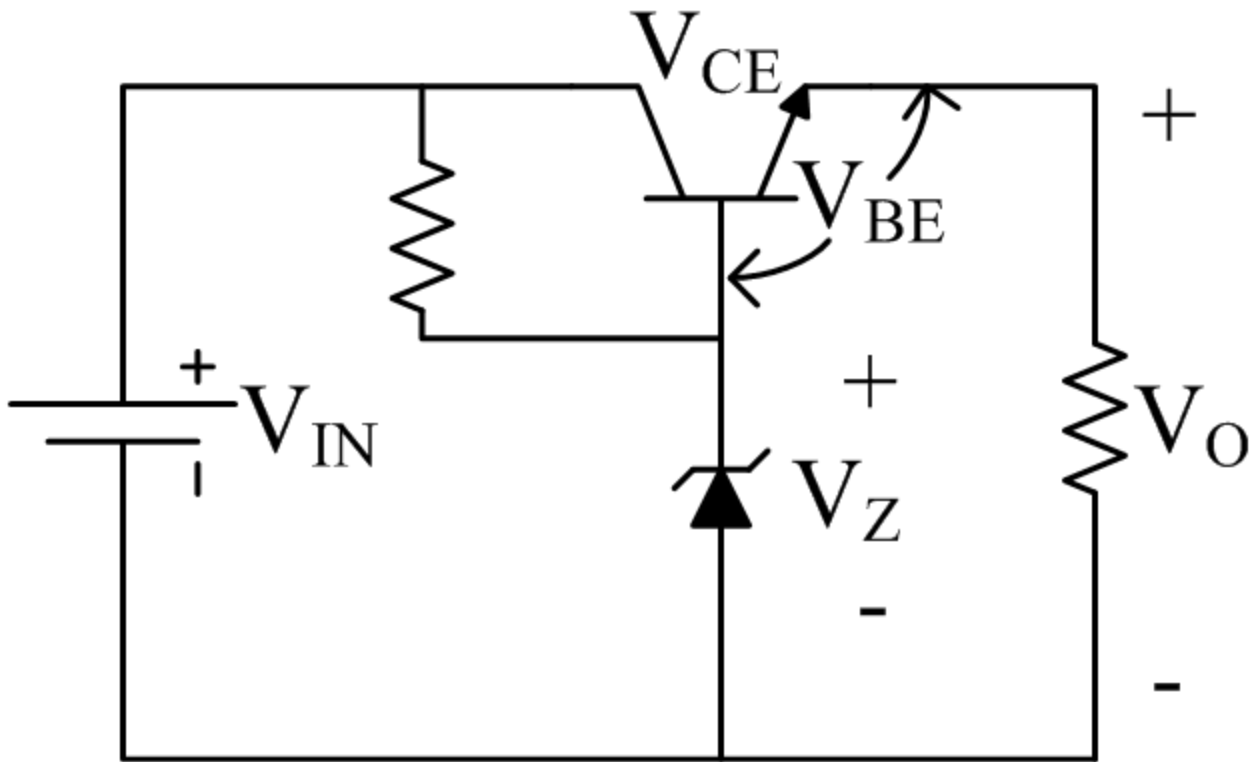
Figure 6-1 A basic linear regulator.

Disadvantages

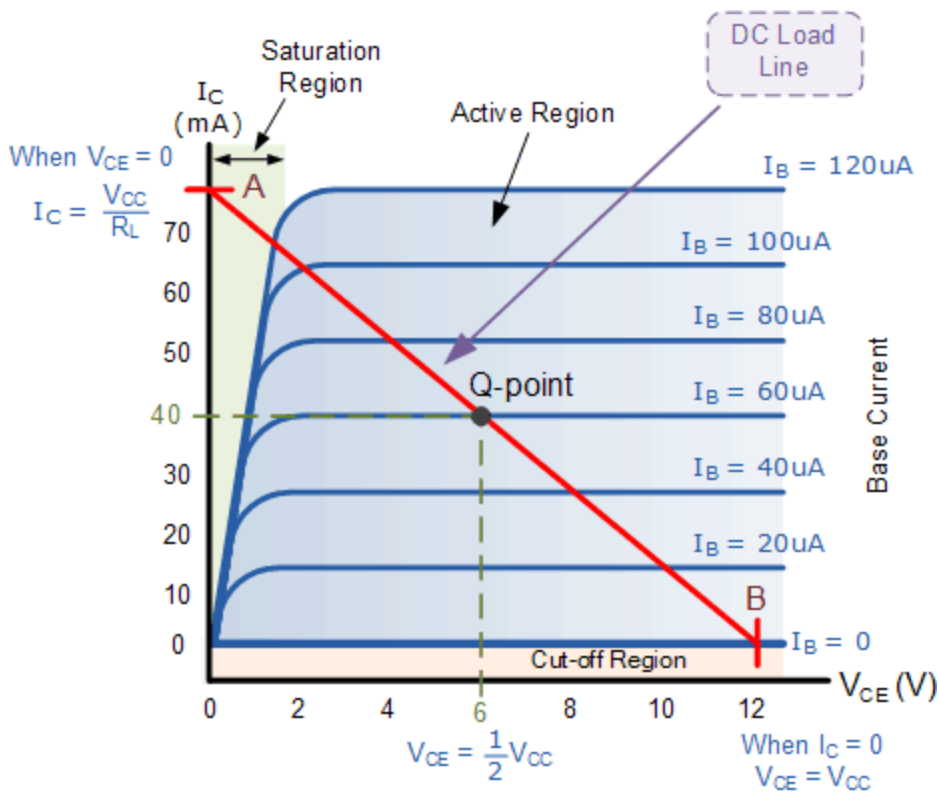
While this may be a simple way of converting a dc supply voltage to a lower dc voltage and regulating the output, the low efficiency of this circuit is a serious drawback for power applications. The power loss in the transistor makes this circuit inefficient. Therefore, the linear voltage regulator is suitable only for low-power applications.

Ref: [Daniel W. Hart - Power Electronics \(2010, McGraw-Hill Science_Engineering_Math\)](#)

Like a transformer, it could be used to step-down or step-up a dc voltage source.



$$V_O = V_{in} - V_{CE}, V_O = V_Z - V_{BE}, V_O = \cos nt - V_{BE}, V_O + V_{BE} = \text{const}$$



if $V_O \uparrow V_{BE}$ must be $\downarrow i_B \downarrow V_{CE} \uparrow$. Since $V_O = V_{in} - V_{CE}$, increase in V_{CE} will decrease $V_{in} - V_{CE}$ and hence maintain $V_O \approx \text{const}$.

if $V_O \downarrow V_{BE}$ must be $\uparrow i_B \uparrow V_{CE} \downarrow$. Since $V_O = V_{in} - V_{CE}$, decrease in V_{CE} will increase $V_{in} - V_{CE}$ and hence maintain $V_O \approx \text{const}$.

DC-DC Switch Mode Converters

- Buck (Step-Down)
- Boost (Step-Up)
- Buck-Boost (Step-Down/Step-Up)
- Cuk (Step-Down/Step-Up)

Buck (Step-Down)

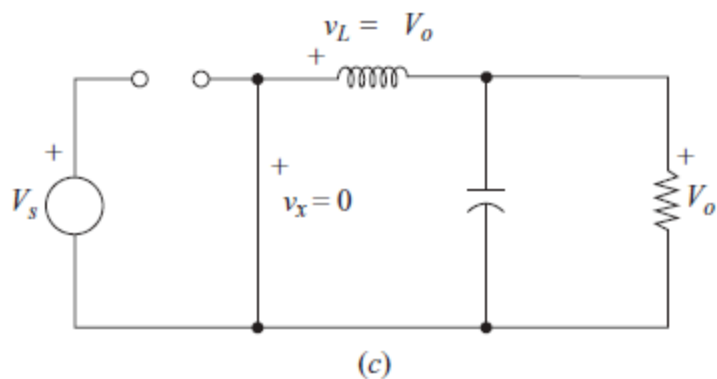
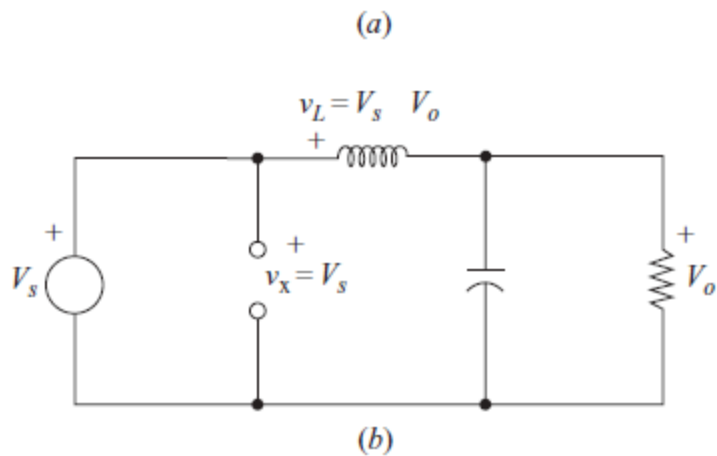
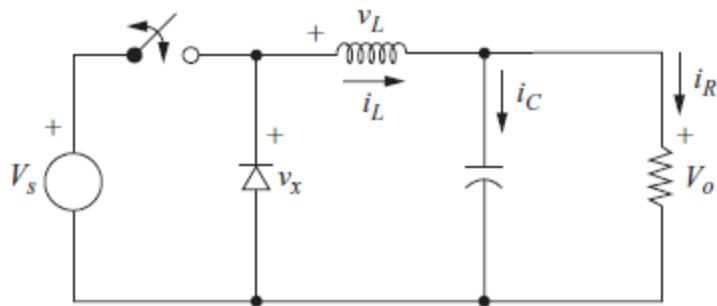
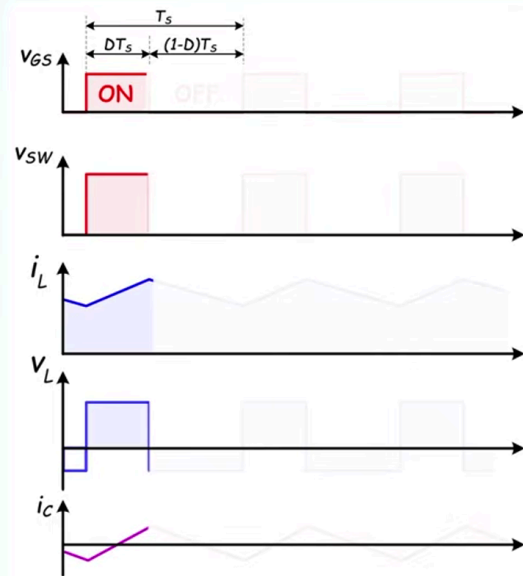
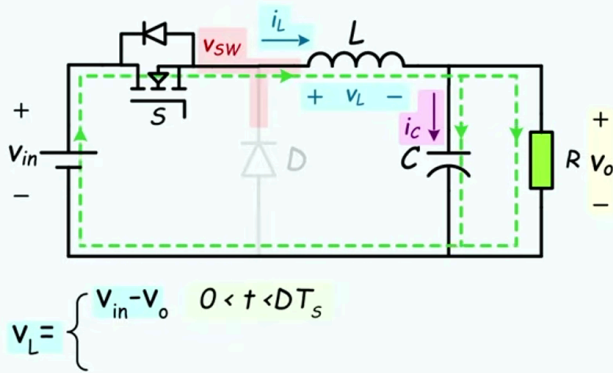


Figure 6-3 (a) Buck dc-dc converter; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

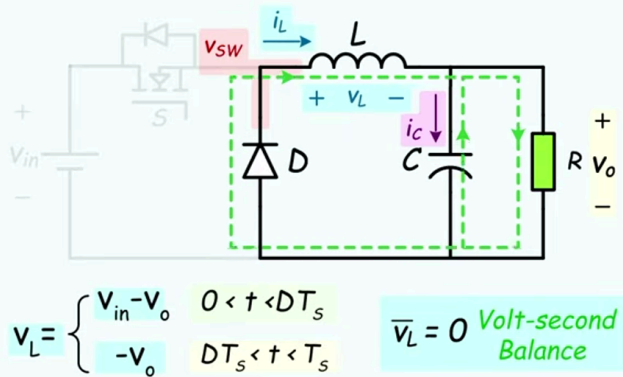
ON

Buck Converter Operation



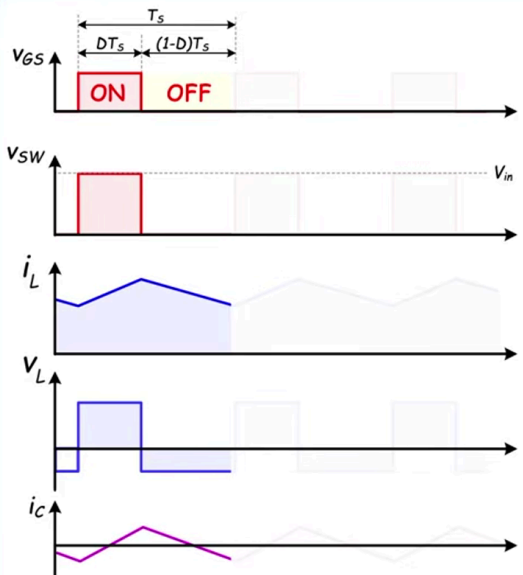
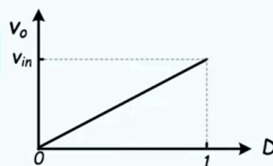
OFF

Buck Converter Operation



$$(V_{in} - V_o) DT_s - V_o (1-D) T_s = 0$$

$$V_o = DV_{in}$$



Boost(Step-Up)

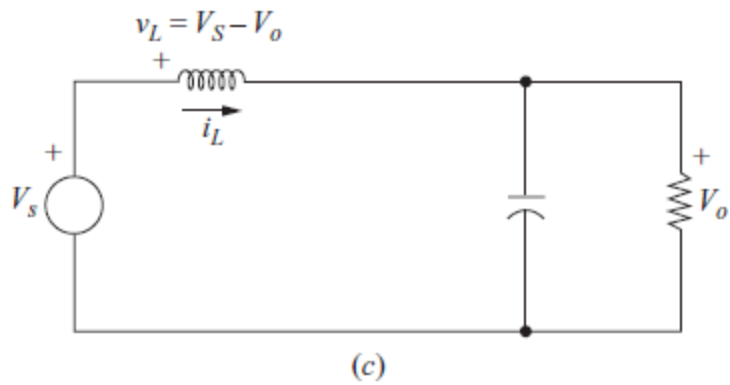
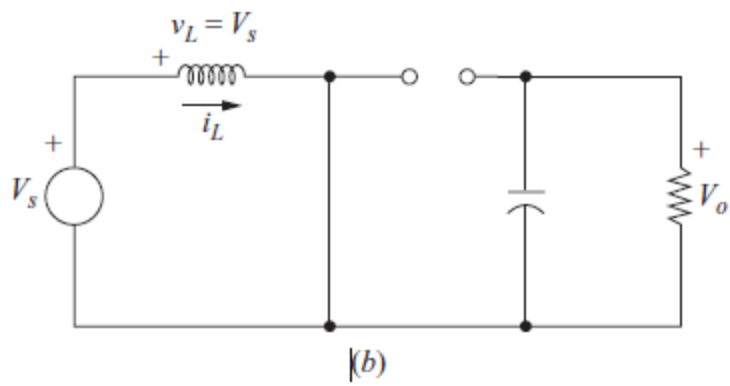
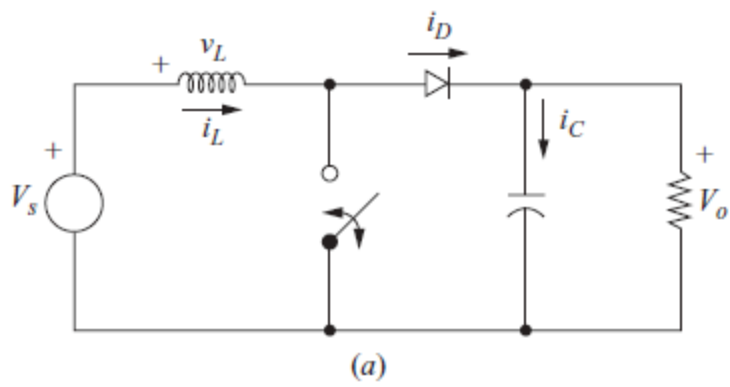
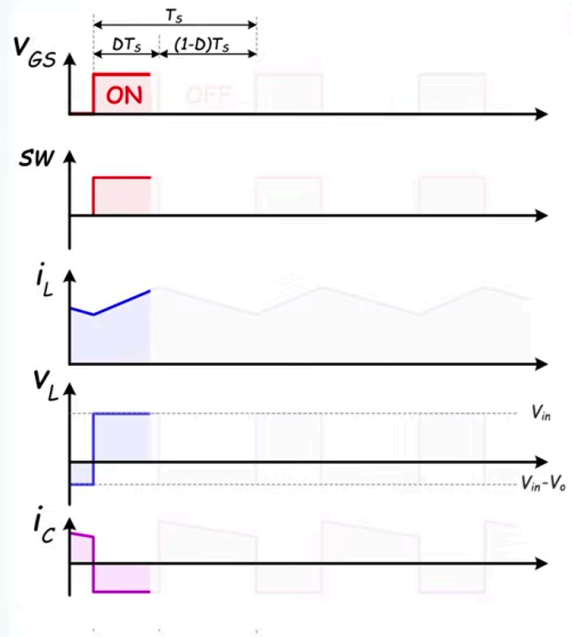
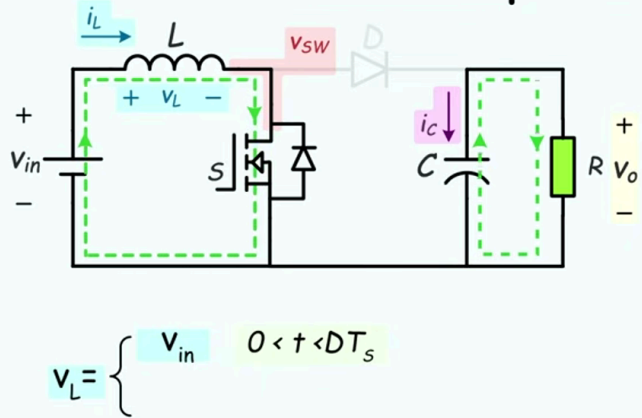


Figure 6-8 The boost converter. (a) Circuit; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

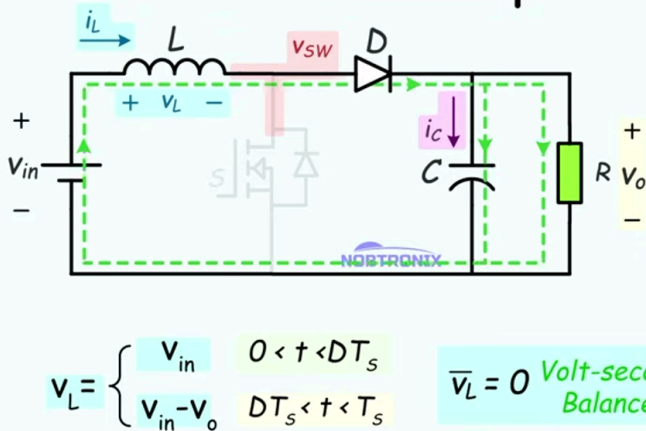
ON

Boost Converter Operation



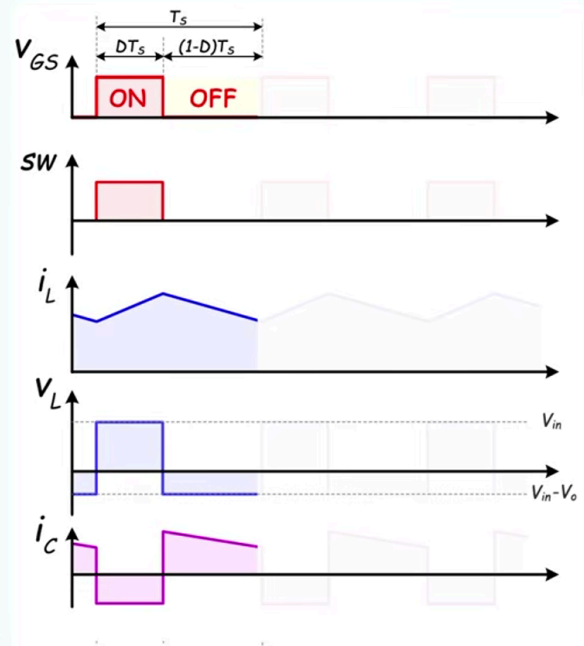
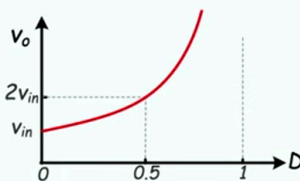
OFF

Boost Converter Operation

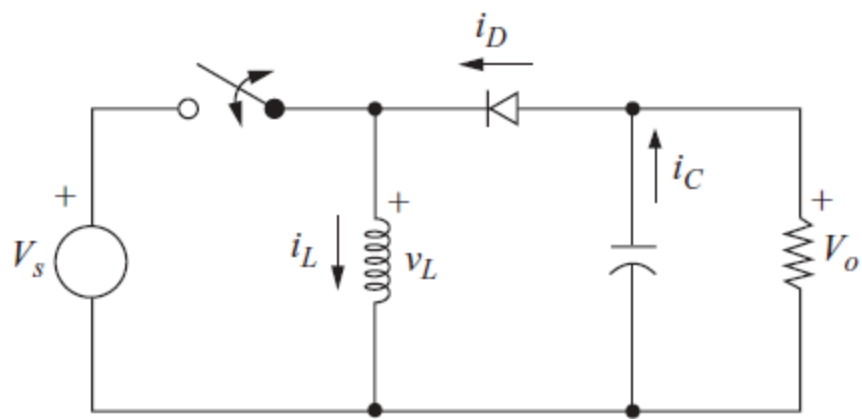


$$V_{in} DT_s + (V_{in} - V_o)(1-D)T_s = 0$$

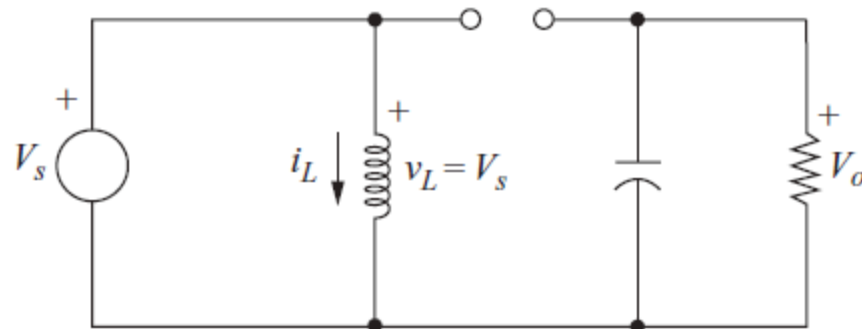
$$V_o = \frac{1}{1-D} V_{in}$$



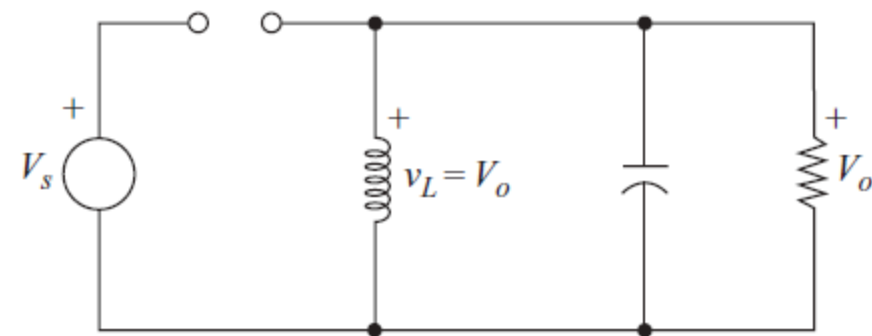
Buck-Boost(Step-down/up)



(a)



(b)

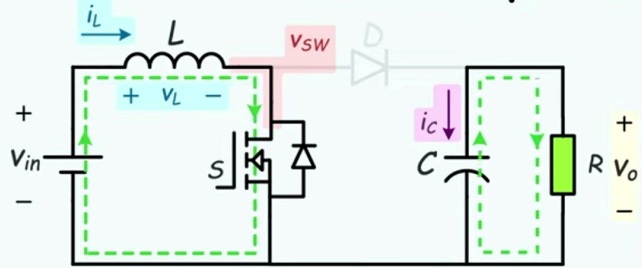


(c)

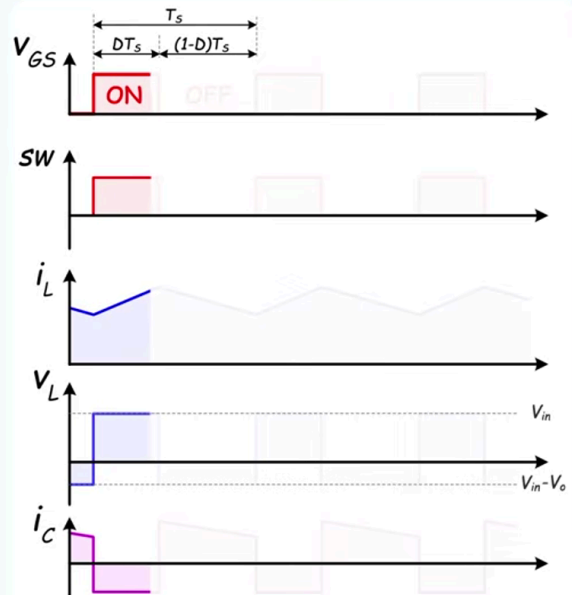
Figure 6-11 Buck-boost converter. (a) Circuit; (b) Equivalent circuit for the switch closed; (c) Equivalent circuit for the switch open.

ON

Boost Converter Operation

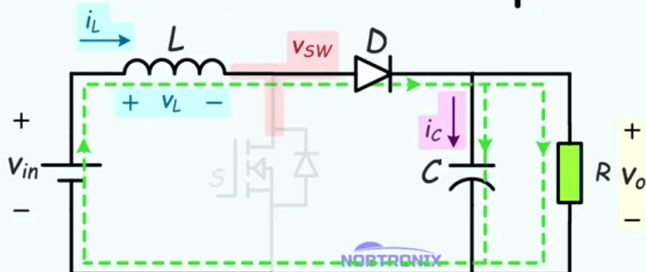


$$v_L = \begin{cases} V_{in} & 0 < t < DT_s \end{cases}$$



OFF

Boost Converter Operation



$$v_L = \begin{cases} V_{in} & 0 < t < DT_s \\ V_{in} - V_o & DT_s < t < T_s \end{cases}$$

$$\bar{v}_L = 0 \text{ Volt-second Balance}$$

$$V_{in} DT_s + (V_{in} - V_o)(1-D)T_s = 0$$

$$V_o = \frac{1}{1-D} V_{in}$$

